Notes: PROBABILITY and PROBABILITY DISTRIBUTIONS

PROBABILITY

Definition: the probability of an event occurring means the likelihood or chance of it occurring.

$$Pr(x) = \frac{\text{favourable no. of cases}}{\text{exhaustive no. of cases}}$$

EXPECTATION

Definition: expectation can be defined as the product of the probability of success and the number of attempts

$$E(x) = \mu = pn = \text{mean}$$

INDEPENDENT EVENTS

When the occurrence of one event does not affect the probability of the occurrence of another, then the events are called independent (a process called drawing 'with <u>replacement</u>')

DEPENDENT EVENTS

Conversely, when the probability of one event occurring does affect the probability of another event occurring, then the events are called dependent events (a process called drawing '<u>without replacement</u>')

eg. bag containing 6 red and 4 black balls $\Rightarrow S = 10$, $r = 6$, $b = 4$

(i) drawing with replacement:

$\Rightarrow$ draw 2 balls in turn

$Pr(1^{st} \text{ ball} = \text{red}) = 6/10$

$Pr(2^{nd} \text{ ball} = \text{red}) = 6/10$

} Indept Event ✓

(ii) without replacement

$Pr(1^{st} R) = 6/10$

$Pr(2^{nd} R) = 5/9$

} dept Events ✓

MUTUALLY EXCLUSIVE EVENTS

Two or more events are called mutually exclusive when the occurrence of one of them excludes the occurrence of the others i.e. when not more than one event can happen at the same time

eg. toss a coin once $\Rightarrow$ H $\Rightarrow$ no tail

toss a die once $\Rightarrow$ 6 shows up $\Rightarrow$ others X

normal birth $\Rightarrow$ boy / girl not both

LAWS OF PROBABILITY

THE ADDITION LAW OF PROBABILITY

This law applies to mutually exclusive events and is usually recognized by the words 'either….or'

It states that when two events are mutually exclusive, if the probability of the first event happening is p_1 and the probability of the second event happening is p_2, the probability of either the first or the second event

happening is
p_1+p_2

THE MULTIPLICATION LAW OF PROBABILITY

This law applies to both dependent and independent events and is usually recognized by the words 'both….and'
It states that when the probability of the first event happening is p_1 and the probability of the second event happening is p_2, the probability of both the first and the second event happening is $p_1 x p_2$

NB: the probability of any event lies within the range $0 \leq P(x) \leq 1$

$Pr(X) = 0 \Rightarrow$ impossible event
$Pr(X) = 1 \Rightarrow$ certain event

QUESTION

Two balls are drawn in turn with replacement from a bag containing 8 red, 15 white, 24 black and 17 orange balls.
Determine the probabilities of having:

a) Two red balls

Exhaustive no. of cases $= 64 = 8+15+24+17$

$Pr(2R) = Pr(R \text{ and } R) = Pr(R) \times Pr(R)$
$= \frac{8}{64} \times \frac{8}{64} = \frac{1}{64}$ ✓

b) A red and a white ball

$Pr(R \text{ and } W) = Pr(R \text{ and } W) \text{ or } Pr(W \text{ and } R)$
$= \left(\frac{8}{64} \times \frac{15}{64}\right) \times 2 = \frac{15}{256}$ ✓

c) No orange balls

$Pr(\text{no } O. \text{ \& no } O.) = \frac{47}{64} \times \frac{47}{64} = \frac{2209}{4096}$ ✓

d) A black and red or a black and orange ball

d) A black and red or a black and orange ball

$$Pr(B \& R \text{ or } B \& O) = \left(\frac{24}{64} \times \frac{8}{64}\right) \times 2 + \left(\frac{24}{64} \times \frac{17}{64}\right) \times 2$$

$$= \frac{75}{256}$$

e) At least one black ball

$$Pr(\text{at least one } B) = Pr(B \& \text{non-}B \text{ or } B \& B)$$

$$= \left(\frac{24}{64} \times \frac{40}{64}\right) \times 2 + \left(\frac{24}{64}\right)^2 = \frac{39}{64}$$

f) At most one orange ball

$$Pr(O \& \text{non } O \text{ or non } O \& \text{no. } O)$$

$$\left(\frac{17}{64} \times \frac{47}{64}\right) \times 2 + \left(\frac{47}{64}\right)^2 = \frac{3807}{4096}$$

g) A white ball on the first draw but the second ball not white

$$Pr(W \& \text{Non-}W) = \frac{15}{64} \times \frac{49}{64} = \frac{735}{4096}$$

ASSGT 1

One bag contains 3 red and 5 black marbles and a second bag contains 4 green and 7 white marbles. One marble is drawn from the first bag and two marbles from the second bag, without replacement.
Determine the probability of having;

a) One red and two white marbles
b) No green marbles
c) Either one black and two green or one black and two white marbles

PROBABILITY DISTRIBUTIONS

Discrete Data ⇒ Counted data
Cont. Data ⇒ Measured dat

3 Prob Distrib's
(i) Binomial } Discrete Prob. distrib
(ii) Poisson }
(iii) Normal ⇒ Continuous Prob. distrib

Binomial ⇒ Small (<30)
Poisson ⇒ large (>30)

THE BINOMIAL PROBABILITY DISTRIBUTION

There are four characteristics that a probability distribution must have in order to be a binomial probability distribution.

Binomial probability distribution results from a procedure described as follows:

1. The procedure has a fixed number of trials [n trials]
2. The trials must be independent.
3. Each trial is in one of two mutually exclusive categories (SUCCESS OR FAILURE)
4. The probabilities remain constant for each trial.

Notations:

$p + q = 1 = \frac{100}{100} = 100\%$

P(success) = P(S) = p probability of success in one of the n trials

P(failure) = P(F) = 1 – p = q probability of failure in one of the n trials

n = fixed number of trials; x = number of successes, where $0 \leq x \leq n$

P(x) = probability of getting exactly x successes among the n trials

$P(x \leq a)$ = probability of getting x-values less than or equal to the value of a.

$P(x \geq a)$ = probability of getting x-values greater than or equal to the value of a.

NOTE: Success (failure) does not necessarily mean good (bad).

One of the four conditions required for a binomial probability distribution is that the trials must be independent. ⇒ Drawing with replacement ✓

If sampling is done with replacement, the trials will be independent.

However, if sampling is done without replacement, the trials are not independent.

When sampling with or without replacement, the trials may be considered to be independent if the sample size is no more than 5% of the population size.

If $n \leq 0.05N$, then the trials may be considered to be independent.

We should be able to calculate binomial probabilities by any of three methods: by formula, from table values, and by using technology.

Formula for Binomial Probabilities:

$Pr(X) = {}^nC_x \, p^x q^{n-x}$

Properties of binomial Prob. Distri

① $\mu = np$, ② $Var = npq$ ③ $\delta = \sqrt{Var}$

Factorial definition: n! = n(n – 1)(n – 2)…2.1; 0! = 1; 1! = 1

by defn

1. QUESTION

The expected number of defective discs in a box is 20 and the variance is 16. Determine the

(i) Probability of selecting a defective disc from the box

(ii) Probability of selecting a non-defective disc from the box

(iii) Number of discs in the box

$E(X) = \mu = 20$ $\quad \delta^2 = 16 = npq$

$= np$ $\quad 16 = 20q$

$q = \frac{16}{} = \frac{4}{}$, $p + q = 1$

$$q = \frac{16}{20} = \frac{4}{5}, \quad p+q=1$$

$$\therefore \; p = 1 - \frac{4}{5} = \frac{1}{5}$$

(i) $Pr(\text{selecting a defective}) = \frac{1}{5} = p$

(ii) $Pr(\text{selecting a non defective}) = \frac{4}{5} = q$

(iii) $np = 20 \Rightarrow n \times \frac{1}{5} = 20 \Rightarrow \underline{\underline{n = 100}}$

2. QUESTION

Some 20 unbiased coins are tossed at once. Determine the probability of having:

(i) Exactly 8 heads

(ii) At least 2 heads

(iii) At most 3 heads

$$P(X) = {}^{n}C_{x}\, p^{x}\, q^{n-x}$$

$n = 20, \; p = q = \frac{1}{2}$

(i) $Pr(X=8) = {}^{20}C_{8} \left(\frac{1}{2}\right)^{8} \left(\frac{1}{2}\right)^{20-8} = {}^{20}C_{8} \left(\frac{1}{2}\right)^{20}$

$= 0.120134353 6$

(ii) $Pr(\text{at least } 2\ H) = 1 - Pr(X = 0, 1)$

$= 1 - {}^{20}C_{0} \left(\frac{1}{2}\right)^{20} - {}^{20}C_{1} \left(\frac{1}{2}\right)^{20}$

$= 0.99997799728$

(iii) $Pr(\text{at most } 3\ H) = Pr(X = 0 \text{ or } 1 \text{ or } 2 \text{ or } 3)$

3. QUESTION

It is expected that 10% of the production from a continuous process will be defective and scrapped. Determine the probability that in a sample of 10 units chosen at random:

(i) Exactly two will be defective

(ii) At most two will be defective;

Use in each case both the binomial distribution and Poisson approximation to the binomial distribution

NB/ When the value of n in a binomial distribution is large and the value of p is very small, the binomial distribution can be approximated by a Poisson distribution.

If n > 20 and np < 5 OR nq < 5 then the Poisson is a good approximation.

ASSGT 2

At a small college, it has been determined that 30% of all students have some type of scholarship support.

(a) If a sample of 10 students is selected at random determine the probability that more than 9 would have some type of scholarship support

(b) Determine the mean number of students having scholarships in a sample of 500 students

THE POISSON PROBABILITY DISTRIBUTION

A discrete random variable X which follows a Poisson distribution has probability density function of the form

$$P(X) = \frac{\lambda^x e^{-\lambda}}{x!}$$

for x = 0, 1, 2, 3, ... to infinity, where $\lambda > 0$ is a positive constant.

We write $X \sim Po(\lambda)$

Note: λ is the parameter of the distribution.

The probability of x successes is $Pr(X) = \frac{\lambda^x e^{-\lambda}}{x!}$

Where e is a mathematical constant = 2.7183; λ is the mean or expected value of the variables.

Expectation and Variance

$E(X) = \text{mean} = \mu = np = \lambda$

If $X \sim Po(\lambda)$, then $E(X) = \lambda$ and $Var(X) = \lambda$

$\delta = \sqrt{Var}$

Poisson Distribution Is an approximation to Binomial Distribution

Since the Poisson distribution arises from the Binomial distribution as a limiting case as n approaches infinity, p tend to 0, in such a way that $np = \lambda$ (a constant), it gives a good approximation to the Binomial distribution for large n and small p.

So, if a random variable X has a binomial distribution given by $X \sim Bin(n,p)$ and n is large and p is small, then X is approximately a Poisson random variable with mean $\lambda = np$ i.e. $X \sim Po(\lambda=np)$

Trivia:

Poisson distribution was derived from the Binomial distribution by Poisson, a 19th century French mathematician Simeon-Denis Poisson (1781-1840). He was also a lecturer, a Professor and examiner who made contributions to mathematical physics

1. QUESTION

It is estimated that 80% of the PC users have installed proprietary operating system. In a group of 1000 PC users, compute the:

(i) Mean

(ii) Standard deviation of the users of proprietary operating system

$\text{mean} = Var = \lambda$

$\delta = \sqrt{Var}$

Solution: $p = 0.8$ $q = 0.2$, $n = 1000$

$\text{mean} = E(X) = np = 1000 \times 0.8 = 800$

$\delta = \sqrt{\lambda} = \sqrt{800}$

2. QUESTION

On average one in 200 cars breaks down on a certain stretch of road per day. Find the probability that on a certain day:

(i) None of a sample of 250 cars breaks down,

(ii) More than 2 of a sample of 300 cars break down. [0.2865047969; 0.1911531695]

3. QUESTION

An insurance company found that 0.01% of the population involved in a certain type of accident each year follows a Poisson probability distribution. If 100 policy holders were randomly selected from the population, determine the probability that not more than two of its clients would be involved in such an accident the following year. (8 marks)

ASSGT 3

A random sample of 800 students in Lwama technical institute produced 80 left-handed students. Calculate the:

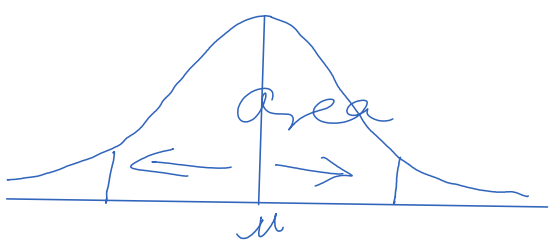
(i) Mean proportion of left-handed students in the institute

(ii) Standard deviation for the number of left-handed students (4 marks)

NORMAL PROBABILITY DISTRIBUTION

The Gaussian or normal probability distribution is the single most important probability distribution.
It was first described by Abraham de Moivre in 1733 but bears the name of Karl Friedrich Gauss, who arrived at this distribution in 1809 when examining the distribution of errors in the measurement of the diameters of lunar craters.
It arises naturally in many other situations (especially the Central Limit Theorem).
Many instances occur where the distributions are continuous, particularly where data relates to measured quantities such as length, time, mass, weight, electric current, temperature, luminous intensity and their derived units.

When a large number of values are taken, of say, length, time, mass, weight, electric current, temperature, luminous intensity or marks gained in an examination, it is found that when the data is plotted as a frequency curve, a curve of similar shape to the normal probability curve is produced in each case.
If any normal probability curve is drawn and the standard deviation, δ, of the data calculated, then the area under the normal curve and the value of the standard deviation are related.



It is found that if a vertical line is drawn to pass through the maximum value of the normal probability curve (the mean value since the curve is symmetrical) and two other vertical lines are drawn at a distance of one standard deviation on either side of this line, then in all cases:

- the area enclosed by the curve and the vertical lines at distance ±1δ from the mean value is about $66\frac{2}{3}$% of the total area under the normal curve

Similarly, it is found that:

- the area enclosed by the curve and the vertical lines at distance ±2δ from the mean value is about 95% of the total area under the normal curve

Finally, it is found that:

- the area enclosed by the curve and the vertical lines at distance ±3δ from the mean value is about $99\frac{3}{4}$% of the total area under the normal curve

Total area under the normal curve is = 100%=1

The relationship a normal probability curve and the area contained between the vertical lines drawn at various

distances between the mean value forms the basis for such techniques such as sampling and quality control.

1. QUESTION

The mean mass of 200 people is 67kg and the standard deviation is 7 kg. Assuming that the masses are normally distributed, determine how many people have a mass:

(i) between 60 and 74kg

(ii) of more than 81kg

(iii) between 53 and 88kg

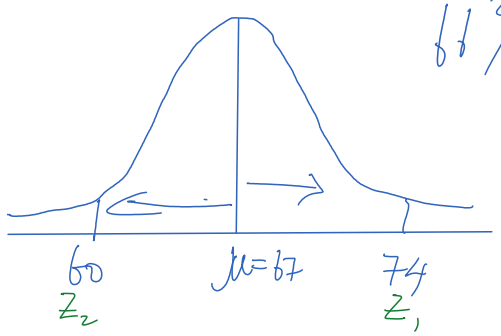
$n = 200$, $\mu = 67\text{kg}$ $\delta = 7\text{kg}$.

$Z_1 = \frac{x-\mu}{\delta} = \frac{74-67}{7} = 1$

$Z_2 = \frac{60-67}{7} = -1$

$66\frac{2}{3}\%$ of $200 =$

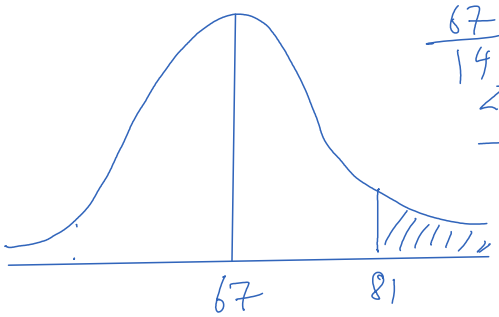
$133.3 \simeq 134$ people



$81 \simeq 2\delta$

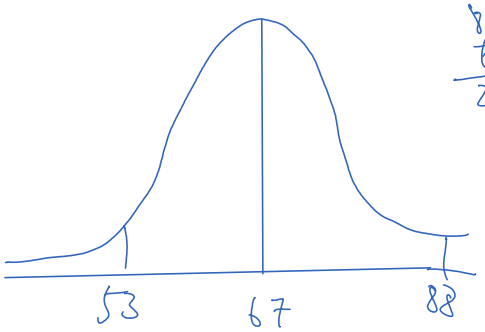
$\begin{array}{r} 81 \\ 67 \\ \hline 14 \end{array}$

$\frac{200 - (95\% \text{ of } 200)}{2} = 5$ ple



$\begin{array}{r} 88 \\ 67 \\ \hline 21 \end{array} \Rightarrow 3\delta$

$\begin{array}{r} 67 \\ 53 \\ \hline 14 \end{array} \Rightarrow 2\delta$



$\frac{1}{2}\left(95\% \text{ of } 200 + 99\frac{3}{4}\% \text{ of } 200\right) \simeq 195$ people

SOLUTION

(i) 134 people

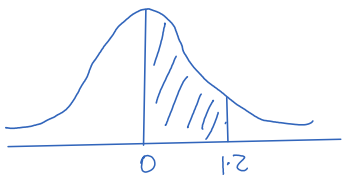
(ii) 5 people

(iii) 195 people

Notation: $X \sim N(\mu, \sigma^2)$
$\mu = E[X]$ = population mean; $\sigma^2 = Var[x]$ = population variance

$Z \sim N(0, 1)$ is the standard normal distribution,
Find the area under the normal curve in each of the following cases:

(i) Between z=0 and z=1.2



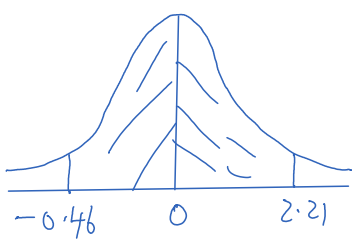
Reqd area = 0.3849

(ii) Between z=-0.68 and z=0

symmetry ⇒ Normal Curve is symmetrical

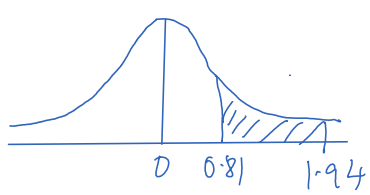
Reqd area = 0.2517

(iii) Between z=-0.46 and z=2.21



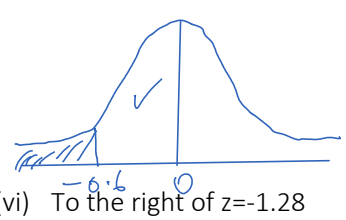
Reqd area: 0.1772 +
0.4864
0.6636

(iv) Between z=0.81 and z=1.94



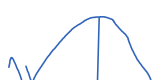
Reqd area: 0.4738
− 0.2910
0.1828

(v) To the left of z=-0.6



Reqd area = 0.5000
− 0.2257
0.2743

(vi) To the right of z=-1.28



Reqd area = 0.5000 +
0.3997

Reqd area = 0.5000 +
0.3997
0.8997

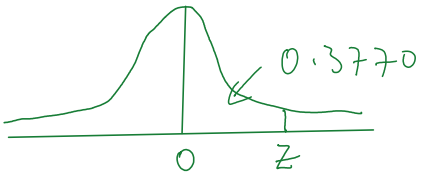
(vii) To the right of z=2.05 and to the left of z=-1.44

Reqd area
0.4251
0.4798 +
0.9049

1 − 0.9049 = 0.0951

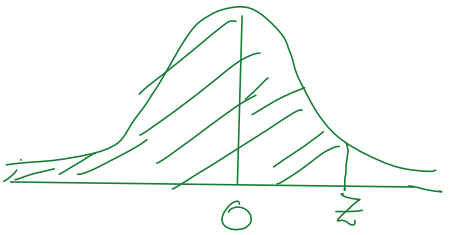
SOLUTION
(i) 0.3849
(ii) 0.2518
(iii) 0.6636
(iv) 0.1828
(v) 0.2742
(vi) 0.8997
(vii) 0.0951

Determine the value of z in each of the following cases:
(i) The area between 0 and z is 0.3770



Z = 1.16

(ii) The area to the left of z is 0.8621



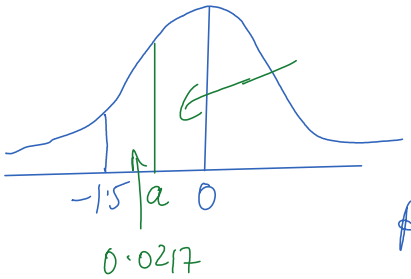
Given that 0.8621 Contains
0.5000
⇒ 0.8621 − 0.5000
= 0.3621 ⇒ from tables
Z = 1.09

(iii) The area between -1.5 and z is 0.0217

0.0217

(iii) The area between -1.5 and z is 0.0217

0.0217



Area $\neq 0.5$

$Z=-1.5 \Rightarrow$ Area

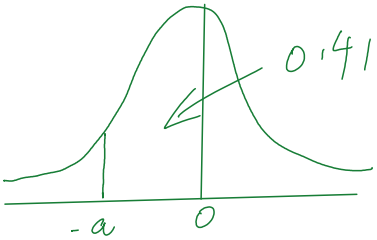
Area btn $Z=0$ & $Z=-1.5$

$= 0.4332$

Btn $Z=a$ & $Z=-1.5$ Area $=0.0217$

$Z=0$ & $Z=-1.5$ Area $=0.4332$

$0.4332-0.0217=0.4115$



$0.4115 \Rightarrow Z=-1.35$

Area btn $Z=0$ & $Z=-1.5$

$= 0.4332$

Sum of Two areas:

$$\begin{array}{r} 0.4332 \\ 0.0217 \\ \hline 0.4549 \end{array}$$

$$\begin{array}{r} 0.4554 \\ 0.4549 \\ \hline 0.00052 \end{array}$$

$$\begin{array}{r} 0.4549 \\ 0.4545 \\ \hline +0.0004 \end{array}$$

$Z=-1.69$

SOLUTION

(i) Z=1.16 or z=-1.16

(ii) Z=1.09

(iii) Case I z=-1.35 Case II z=-1.69

2. QUESTION

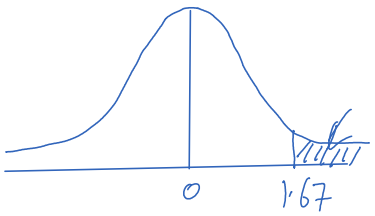
The time taken to learn the standing orders by members of parliament is normally distributed with a mean of 80 hours and a standard deviation of 6 hours. If a random sample of 16 members is selected, find the probability that the mean time to learn the standing orders will be more than 90 hours.

probability that the mean time to learn the standing orders will be more than 90 hours.

$\mu = 80 \quad \delta = 6 \quad Z = \frac{X-\mu}{\delta}$

$Pr(X > 90)$

$= \frac{90-80}{6} = 1.67$



Reqd area $= 0.5000 - 0.4525 = 0.0475$

Prob. $= 0.0475$

3. QUESTION

The life of a computer component is normally distributed with mean μ and standard deviation δ. The probability that the component lasts less than 1980 hours is 0.121 and the probability that the computer component lasts more than 2060 hours is 0.209. Determine the:

(i) Mean life time μ

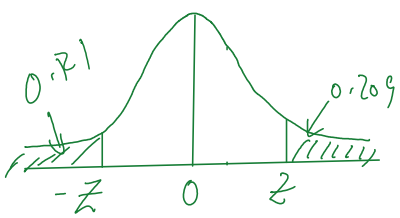
(ii) Standard deviation δ of the lifetimes

(iii) Probability that the component will last longer than 2010 hours

μ & δ

$Pr(X < 1980) = 0.121$

$Pr(X > 2060) = 0.209$



$Z = \frac{1980-\mu}{\delta} = -1.17$

$\frac{2060-\mu}{\delta} = 0.81$

$1980 - \mu = -1.17\delta$

$2060 - \mu = 0.81\delta$

$\Rightarrow \mu - 1.17\delta = 1980$

$\mu + 0.81\delta = 2060$

$\mu = 2027.\dot{2}\dot{7}$

$\delta = 40.\dot{4}\dot{0}$

ASSGT 4

The marks of 500 candidates in an examination are normally distributed with mean of 45 and a standard deviation of 5 marks. Given that the pass mark is 40, estimate the number of candidates who passed the exam. If 4% of the candidates obtained a distinction by scoring X marks or more, estimate the value of X. Estimate the interquartile range of this distribution.

5. QUESTION

A random variable X follows a normal distribution with mean μ and standard deviation δ. If 12.3% of the distribution is greater than 60 and 5.89% is less than 40, find μ and δ.

6. QUESTION

(a) Explain five important uses of normal probability distribution in statistics

(b) On average one in 200 cars breaks down on a certain stretch of a road per day. Find the probability that out of a sample of 250 car selected at random;

(i) None breaks down

(ii) More than two break down

SOLUTION

(a) Importance of the normal distribution:

(i) Frequency distributions of many physical characteristics e.g. height, weight etc., often have a shape of the normal curve

(ii) It is useful as an approximation to various other probability distributions

(iii) By virtue of the "Central Limits Theorem" the distribution of the means of the samples taken from any population, which need not be normal, tend towards the normal distribution if the sample is large

(iv) The normal distribution has extensive uses in the sampling theory

(v) The normal distribution has a wide use in testing statistical hypothesis and test of significance

(any 5@2 marks): (10 marks)

(b) n=250, P=1/200, Q=199/200

(i) μ=np=1.25

(ii) $Pr(X) = \frac{\lambda^x e^{-\lambda}}{x!} = \frac{1.25^0 e^{-1.25}}{0!} =$ ($\lambda = \mu$)

$Pr(X = \text{more than } 2) = 1 - Pr(X = 0 \text{ or } 1 \text{ or } 2)$

Required probability=0.3686

(iii)

Required probability=0.5000-0.2486=0.2514